Cell Biology: The Structure and Function of Cells

Chapter 1: Introduction to Cells

The Discovery of Cells

The cell is the basic unit of life. All living organisms are composed of one or more cells. The discovery and study of cells has been fundamental to understanding biology.

Cytology - The branch of biology concerned with the study of cells, their structure, function, and chemistry.

Cell Theory - A fundamental principle in biology stating that:
1. All living organisms are composed of one or more cells
2. The cell is the basic unit of structure and organization
3. All cells arise from pre-existing cells

Microscope - An instrument that uses lenses to magnify small objects. Robert Hooke first observed cells using a simple microscope in 1665.

Resolution - The ability to distinguish between two closely spaced objects. Modern electron microscopes have much greater resolution than light microscopes.

Types of Cells

Prokaryotic cells - Simple cells without a membrane-bound nucleus. Bacteria and archaea are prokaryotes. These cells are typically smaller (1-10 μm) and have their genetic material in a nucleoid region.

Eukaryotic cells - Complex cells with a membrane-bound nucleus and specialized organelles. Animals, plants, fungi, and protists are eukaryotes. These cells are larger (10-100 μm) and more structurally complex.

Unicellular - Organisms consisting of a single cell, such as bacteria or amoeba.

Multicellular - Organisms composed of many cells that work together, such as humans or trees.

Chapter 2: Cell Structures and Organelles

The Plasma Membrane

Plasma membrane - Also called the cell membrane, this is a selectively permeable barrier that separates the cell's interior from its external environment. It consists of a phospholipid bilayer with embedded proteins.

Phospholipid bilayer - A double layer of phospholipid molecules that forms the basic structure of cell membranes. The hydrophobic tails face inward, while hydrophilic heads face outward.

Selective permeability - The property of membranes that allows some substances to pass while blocking others. This is crucial for maintaining cellular homeostasis.

The Nucleus

Nucleus - The control center of eukaryotic cells, containing the cell's genetic material (DNA). It is surrounded by a double membrane called the nuclear envelope.

Nuclear envelope - A double membrane with pores that regulates the passage of materials between the nucleus and cytoplasm.

Chromatin - The complex of DNA and proteins found in the nucleus. During cell division, chromatin condenses into visible chromosomes.

Nucleolus - A dense region within the nucleus where ribosomal RNA is produced.

Chromosomes - Structures made of condensed chromatin that carry genetic information in the form of genes.

The Cytoplasm

Cytoplasm - The gel-like substance filling the cell between the plasma membrane and nucleus. It contains organelles, cytosol, and the cytoskeleton.

Cytosol - The liquid portion of the cytoplasm, composed mainly of water with dissolved ions, molecules, and proteins.

Cytoskeleton - A network of protein filaments providing structural support, enabling cell movement, and helping transport materials within the cell.

Energy-Producing Organelles

Mitochondria (singular: mitochondrion) - Organelles responsible for cellular respiration, converting nutrients into ATP (energy). Often called the "powerhouse of the cell." They have their own DNA and double membrane.

ATP (Adenosine Triphosphate) - The primary energy currency of cells. Energy released from breaking ATP bonds powers most cellular processes.

Cellular respiration - The process by which cells break down glucose and other organic molecules to produce ATP, consuming oxygen and producing carbon dioxide.

Chloroplasts - Organelles in plant cells and some protists where photosynthesis occurs. They contain chlorophyll and convert light energy into chemical energy (glucose).

Protein Manufacturing and Processing

Ribosomes - Small structures composed of RNA and proteins where protein synthesis occurs. They can be free-floating in the cytoplasm or attached to the endoplasmic reticulum.

Endoplasmic Reticulum (ER) - A network of membranous tubules and sacs involved in protein and lipid synthesis.

Rough ER - ER studded with ribosomes; primarily involved in protein synthesis and modification.

Smooth ER - ER without ribosomes; involved in lipid synthesis, carbohydrate metabolism, and detoxification.

Golgi apparatus - Also called Golgi complex or Golgi body, this organelle modifies, packages, and sorts proteins and lipids for transport to their final destinations.

Vesicles - Small membrane-bound sacs that transport materials within cells or to the cell membrane for secretion.

Storage and Waste Management

Vacuoles - Membrane-bound sacs used for storage. Plant cells typically have one large central vacuole that stores water, nutrients, and waste products, helping maintain turgor pressure.

Lysosomes - Organelles containing digestive enzymes that break down waste materials, cellular debris, and foreign substances. Sometimes called the "recycling centers" of the cell.

Peroxisomes - Organelles that break down fatty acids and detoxify harmful substances, producing hydrogen peroxide as a byproduct (which they then break down).

Chapter 3: Specialized Structures

Plant Cell Structures

Cell wall - A rigid structure outside the plasma membrane in plant cells, fungi, and some protists. In plants, it's primarily made of cellulose and provides structural support.

Cellulose - A complex carbohydrate that forms the primary component of plant cell walls.

Plasmodesmata - Channels through plant cell walls that allow direct cell-to-cell communication and transport.

Movement Structures

Cilia (singular: cilium) - Short, numerous hair-like projections that move in coordinated waves to propel cells or move substances across cell surfaces.

Flagella (singular: flagellum) - Long, whip-like appendages that propel cells through fluid environments. Sperm cells use flagella for movement.

Chapter 4: Cell Transport

Passive Transport

Diffusion - The movement of molecules from areas of high concentration to low concentration, requiring no energy input.

Osmosis - The diffusion of water across a selectively permeable membrane from areas of low solute concentration to high solute concentration.

Facilitated diffusion - The passive movement of molecules across membranes with the help of transport proteins.

Active Transport

Active transport - The movement of molecules against their concentration gradient (from low to high concentration), requiring energy (ATP).

Endocytosis - The process by which cells engulf external materials by forming vesicles from the plasma membrane.

Exocytosis - The process by which cells expel materials in vesicles that fuse with the plasma membrane.

Phagocytosis - A type of endocytosis where cells engulf large particles or other cells ("cell eating").

Pinocytosis - A type of endocytosis where cells take in liquid and small dissolved molecules ("cell drinking").

Conclusion

Understanding cell structure is fundamental to understanding how living organisms function. Each organelle has specialized functions that contribute to the cell's overall survival and operation. The coordination between these structures allows cells to grow, reproduce, respond to their environment, and maintain homeostasis. Modern cell biology continues to reveal new insights into cellular mechanisms, advancing medicine, biotechnology, and our understanding of life itself.